VIKRANTH VAKATI, M.S.

Systems Engineer | Boston, MA | U.S. Citizen

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EXPERIENCE

Systems Engineer | Bodkin Design & Engineering | Newton, MA — May 2025 - Present

Electro-optical imaging solutions design for industrial and research communities

- Led development of software for system control and image processing using Python
- Collaborated in a cross-functional team to integrate software with hardware components
- Enhanced system performance by evaluating and implementing emerging technologies

Electrical Test Engineer | Delta Magnetics & Controls | Holbrook, MA — Jan 2021 - Jan 2023

Control panel design and fabrication for industrial process automation

- Wired and assembled control panels following industry standard IEC schematics
- Conducted visual and point-to-point testing to verify system functionality
- Deployed over 1000 control panels adhering to operational requirements

PROJECTS

Spacecraft Materials Testing | Image Processing, Data Management | Python, Numpy — Apr 2027

In-situ imaging system to characterize material degradation in a simulated space environment

- Composed wiring schematics compliant with class 3B laser safety requirements
- Designed software for multithreaded motor control and image acquisition
- Generated BRDF measurements with improved SNR for use in 3D simulations

Autonomous Inspection Drone | Robotics, Machine Learning | Python, YOLO26, MAVSDK — Jan 2027

Aerial imaging system to detect and monitor corrosion over time on galvanized steel support cables

- Developed a closed-loop tracking algorithm to maintain precise drone alignment along a cable
- Leveraged machine learning and sensor fusion to reliably track cables under dynamic conditions
- Improved inspection accuracy and repeatability, enabling long-term condition monitoring

Image Segmentation Tool | Image Processing, UI/UX | Python, OpenCV — Apr 2026

Interactive application to rapidly prototype and analyze classical image segmentation algorithms

- Created a GUI to convert between color spaces and tune real-time camera input
- Applied per channel thresholding, histogram equalization, and morphological filtering
- Streamlined selection of color spaces and thresholding of classical image segmentation

Blackbody Calculator | Mathematical Modeling, UI/UX | JavaScript, HTML/CSS — Jul 2025

Web tool to calculate and visualize spectral radiance and emittance | bit.ly/4uXHdIp

- Implemented models based on first-principle physics laws and validated results
- Deployed a mobile-responsive UI with intuitive controls and real-time updates
- Improved response time compared to conventional tools through lazy data loading

EDUCATION

Wentworth Institute of Technology | Boston, MA

- M.S. in Computer Engineering, Concentration in Internet of Things — May 2024
- B.S. in Computer Engineering, Minors in Electrical Engineering and Applied Math — Dec 2022

SKILLS

Languages: Python, JavaScript, HTML/CSS, C/C++, Bash, MATLAB, SQL
Tools: VS Code, Numpy, OpenCV, YOLO26, SAM3, ImageJ, Windows, Linux, Excel, Git, LabView, MultiSim
Equipment: Multimeter, Oscilloscope, Spectral Cameras, Monochromator, Microcontrollers, SBCs
Communication Protocols: I2C, SPI, UART, RS232, RS485, Wi-Fi, MAVLink, Modbus, LoRa, BLE, Ethernet